

Sales Order Fulfillment Analysis – Summary

Objective

The objective of this analysis was to evaluate open sales orders against available inventory and determine the appropriate action for each order—whether it can be fulfilled, transferred, replenished, or canceled. The solution integrates Sales Orders, On-Hand Inventory, Kit Components, and Open Purchase Orders to support accurate and practical decision-making.

Of 5,759 sales order lines analyzed: 4,822 require no action, 184 are recommended for cancellation, 81 require a new PO, 61 can be resolved via transfer, and 611 have an existing PO to monitor.

Approach

The analysis follows a structured workflow:

- Checked inventory at the requested location to determine immediate fulfillment
- Evaluated inventory across all locations to identify transfer opportunities
- Applied item status rules, ensuring Discontinued and Sell Down items are always canceled
- Reviewed open purchase orders to avoid duplicate procurement
- Evaluated kit/package items at the component level

Decision Logic

Each order was assigned a Final Action based on:

- **C – Cancel Order**
For Discontinued or Sell Down items, regardless of inventory
- **No Action Needed**
If sufficient inventory exists at the requested location (“**Ready to Fulfill**” state)
- **B – Transfer Inventory**
If inventory exists at another location
- **A – Place Purchase Order**
If inventory is unavailable and the item is eligible
- **Existing PO – Monitor**
If an open purchase order already exists for the item (To avoid over-stocking)

Key Components

1. Sales Order Fulfillment:

Main working sheet with:

- Inventory matching (Item + Location key)
- Shortage calculation
- Item status validation with Recommendations
- Final Action and Notes (used notes for detailed explanations)

2. Pivot Table – Fulfillment Status

Provides a high-level summary of:

- Demand vs available inventory
- Location-wise fulfillment status
- Added slicers to filter required data for a quick summary view.

3. Order Lookup & Validation Tool

Interactive tool to:

- Retrieve order details using Document Number
- Display fulfillment status and recommendations

4. Kit Component Logic

Kit feasibility is determined by:

- Breaking kits into components
- Calculating availability per component
- Using the minimum available component to determine fulfillment

For kit/package items, fulfillment is determined at the component level rather than at the kit SKU level. In several cases, the kit shows zero on-hand inventory, but all required components are available. Therefore, these are correctly marked as “No Action Needed.”

5. Open Purchase Orders

Used to:

- Identify incoming inventory
- Prevent duplicate purchase order recommendations

Data Observations

- Identified **71 duplicate rows** in Sales Orders with no unique identifier
- Open Purchase Orders contained inconsistencies such as missing and duplicate entries
- These issues were retained as they did not impact calculations, but would require cleaning in production

Future Enhancements

- Standardize naming conventions for Final Action using shorter, business-aligned codes (e.g., A, B, C, A1)
- Develop an additional SKU-based lookup tool along with a component-level breakdown for kit items.
- Introducing data cleaning and validation processes
- Add timestamps to distinguish duplicate or repeated orders in the SO sheet.
- Implement reorder alerts for proactive inventory management
- Build a dashboard for visual insights, including the count of Final Actions
- Add a glossary sheet for action definitions

Summary

This solution provides a structured and practical approach to sales order fulfillment using inventory, purchase orders, and kit component data. It combines analytical logic with usability features such as pivot summaries and a lookup tool, enabling clear, accurate, and real-time decision-making.

Tools used: Excel (XLOOKUP, SUMIF/SUMIFS, MINIFS, nested IF logic, FILTER, Pivot Tables, VBA macros for the Refresh button, and a dynamic validation tool). Claude AI and GPT-4.0 were used as thinking partners to identify edge cases and structure the written analysis.